🍎 Swift Complete Introduction Guide

A beginner-friendly guide to Apple's powerful programming language.

📄 What is Swift?

Swift is a **modern programming language** developed by Apple for building apps on:

- iOS 📱
- macOS 💻
- watchOS ⌚
- tvOS 📺

In simple terms:

Swift = Fast + Safe + Modern App Development

🕰 History of Swift

Year	Event
2014	Swift introduced by Apple
Creator	Chris Lattner
Developed By	Apple Inc.

Swift replaced Objective-C as the primary language for Apple development.

⭐ Why Use Swift?

Key Benefits

✓ Fast performance
✓ Easy to learn & readable
✓ Safe (prevents common errors)

✓ Modern syntax
✓ Strong Apple ecosystem support

⚡ Core Features of Swift

1. Type Safety

Prevents incorrect data usage.

2. Optionals

Handle missing values safely.

3. Closures

Powerful function-like blocks.

4. Automatic Memory Management

No need to manage memory manually.

5. Fast Performance

Comparable to C-based languages.

🛠 Installing Swift

For macOS:

- Install Xcode from App Store

Verify Installation:

```
swift --version
```

💡 Basic Syntax Example

```
var message: String = "Hello Swift"
print(message)
```

Output:

```
Hello Swift
```

□ Data Types in Swift

Type	Description
Int	Integer values
Double	Decimal numbers
String	Text
Bool	true / false
Optional	Value or nil

Example

```
var age: Int = 25
var name: String = "John"
var isStudent: Bool = true
```

🔧 Functions in Swift

```
func add(a: Int, b: Int) -> Int {
    return a + b
}
```

Usage:

```
print(add(a: 10, b: 5))
```

□ Optionals (Important Feature)

Optionals allow variables to have no value.

```
var name: String? = nil
```

Unwrapping:

```
if let safeName = name {
```

```
    print(safeName)
}
```

Classes in Swift

```
class Student {
    var name: String

    init(name: String) {
        self.name = name
    }

    func greet() {
        print("Hello \(name)")
    }
}
```

Arrays

```
var numbers: [Int] = [1, 2, 3, 4]
```

Dictionaries

```
var student = ["name": "John", "age": "25"]
```

Loops in Swift

```
for i in 1...5 {
    print(i)
}
```

Conditionals

```
let age = 18

if age >= 18 {
    print("Adult")
} else {
    print("Minor")
```

```
}
```

❁ Swift for App Development

Swift is mainly used for:

- iPhone apps 📱
- iPad apps
- macOS software
- Apple Watch apps

Popular tools:

- Xcode
- SwiftUI
- UIKit

□ Example Program

```
func greet(name: String) -> String {
    return "Hello \(name)"
}

print(greet(name: "John"))
```

Output:

```
Hello John
```

□ Advantages of Swift

✓ Fast and efficient
✓ Safe and secure
✓ Easy to read syntax
✓ Strong Apple support
✓ Modern language features

● Disadvantages of Swift

• Limited to Apple ecosystem
• Smaller community than JavaScript
• Frequent updates

🏆 Best Practices

✓ Use optionals safely
✓ Write clean and readable code
✓ Use Swift naming conventions
✓ Keep functions small and reusable
✓ Use SwiftUI for modern apps

📌 Quick Summary

Swift helps developers:

- Build iOS apps easily
- Write safer code
- Develop high-performance apps

✨ Tips to Make Your PDF Look Professional

• Use Apple-style clean design 🍎
• Add icons and section dividers
• Highlight code blocks
• Use fonts like:

- San Francisco (Apple style)
- Inter
- Roboto

Recommended Structure:

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